

Generation Interconnection Request System Impact Study Report

For Queue Project Q17-0819

Buxton 138 kV
July 21, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0819. The Interconnection Customer proposes a 273.2 MW natural gas combined cycle facility interconnecting at the Buxton 138 kV substation in the Pennsylvania area. The request was received on July 02, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 245 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$53,154,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
VOLCANO - WALNUT CREEK 138.0 kV	CORONET - FT HMBUG 500.0 kV line	250.7	240	31.6
BUXTON SUBSTATION - NORTHSIDE DRIVE 138.0 kV	CORONET - FT HUMBUG 500.0 kV	216.8	149	49.3
NRTHSIDE DRIVE - VOLCANO SUBSTATION 138.0 kV	CORONET - FT HUMBUG 500.0 kV	177.5	180	49.3
FT HUMBUG 345/138 kV Autotransformer	FORT HUMBUG - W CAMPUS 345.0 kV	134.3	356	31.2
MIDWIND - HIGH SHELDON 138.0 kV	BUXTON - NORTHSIDE DRIVE SUBSTATION 138.0 kV	131.8	151	6.7
BUXTON - YEAGERTOWN 138.0 kV L	BUXTON - NORTHSIDE DRIVE 138.0 kV	129.6	206	100.0
MONTOUR STA SUBSTATION 500/345kV TX	SUNBURY 500/345kV TX	119.3	883	5.0
MORGANVILLE - IRONWOOD LLC 345.0 kV Line	WIWAM - UPSCO SUBSTATION 345.0 kV	107.7	1,106	6.4
MIDWIND SUBSTATION - SILVERTON 138.0 kV	BUXTON SUBSTATION - NORTHSIDE DRIVE 138.0 kV	105.9	199	6.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VOLCANO - WALNUT CREEK 138.0 kV	Rebuild line with higher-capacity conductor	\$9,249,000	100% (\$9,249,000)
NORTHSIDE DRIVE - BUXTON SUBSTATION 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,949,000	100% (\$5,949,000)
NORTHSIDE RIVE - VOLCANO 138.0 kV L	Rebuild line with higher-capacity conductor	\$6,241,000	100% (\$6,241,000)
FT HUMBUG 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$15,927,000	89% (\$14,106,000)
MIDWIND - HIH SHELDON 138.0 kV	Rebuild line with higher-capacity conductor	\$2,959,000	80% (\$2,362,000)
BUXTON - YEAGERTOWN 138.0 kV L	Rebuild line with higher-capacity conductor	\$8,081,000	100% (\$8,081,000)
MONTOUR STA SUBSTATION 500/345kV TX	Replace transformer with higher-rated unit	\$31,111,000	16% (\$4,895,000)
IRONWOOD LLC - MORGANVILLE 345.0 kV	Reconductor line	\$5,771,000	9% (\$521,000)
MIDWIND SUBSTATION - SILVERTON 138.0 kV	Reconductor line	\$2,192,000	80% (\$1,750,000)
	Total allocated to project		\$53,154,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.